INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA
Term – I ,First Semester Examination, September- 2026
Subject Code: BCACC301
Subject Name: Design and Analysis of Algorithms

Full Marks: 30 Set No:1 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2×5)

1. Define an algorithm. Mention any two characteristics of a good algorithm. **2**

2. Differentiate between a priori analysis and a posteriori analysis. **2**

3. What is meant by the growth rate of a function? **2**

4. Write the recurrence relation for the following recursive function: **2**

```
FUN(n)
{
    if n <= 1
        return 1;
    return FUN(n-1) + n;
}
```

5. What is the main difference between Selection Sort and Insertion Sort? **2**

6. State the condition under which Binary Search can be applied to an array. **2**

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5×2)

7. Explain performance analysis of an algorithms with a suitable example. **5**

8. Compare merge sort and quick sort based on: **5**

a. Divide operation
b. Combine operation
c. Best-case complexity
d. Worst-case complexity
e. Auxiliary space
f. Stability

9. Consider the merge sort recurrence: **5**

$$T(n) = 2T(n/2) + n$$

Use expansion or substitution to show that its complexity is O(n log n)

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10×1)

10. Define an algorithm and explain its essential characteristics in detail. Illustrate each characteristic using a suitable algorithm or real-life example. **10**

11. Write and explain the quick sort algorithm. Apply it to the following array using the first element as the pivot: **10**

[40, 20, 60, 10, 50, 30, 70]

Show every major partitioning step and write the final sorted array.

…END…